

Karuna Venkatesh

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EDUCATION

New York University, Tandon School of Engineering

MS in Computer Engineering, Machine Learning concentration (GPA: 3.61)

Brooklyn, NY

Aug 2025 - May 2027

Dayananda Sagar College of Engineering

BE in Electronics and Telecommunication Engineering

Bangalore, India

Jun 2025

EXPERIENCE

Mattar Lab, New York University

Graduate Researcher

Brooklyn, NY

Jun 2026 - Aug 2026

- Studied how recurrent neural network model capacity shapes model behavior, designing and running a controlled experiment across 8 model capacity conditions and 40 trained models to measure how adversarial attack transferability changes as networks scale in computational models of human decision-making.
- Identified and resolved 8 root-cause issues affecting experimental validity, including a model capacity mismatch and a random seed reproducibility issue, ensuring results reflected genuine differences between models rather than implementation errors.
- Analyzed experimental results across 40 evaluation runs using confidence intervals to characterize the relationship between model capacity and adversarial transfer effect, informing next steps for the research direction.

Consolewaves Pvt Ltd

Front-End Development & UI/UX Intern

Feb 2025 - May 2025

- Built a quiz interface in Figma and JavaScript; ran user testing, identified failure modes, and iterated to improve accessibility.

Beekoder Technologies Pvt Ltd

Full-Stack Web Development Intern

Oct 2023 - Nov 2023

- Architected a restaurant website using Bootstrap and SQL; managed production deployment and resolved critical bugs.

PROJECTS

Parameter-Efficient Fine-Tuning of Vision-Language and Language Models (LoRA & QLoRA)

Python, PyTorch, HuggingFace, LoRA/QLoRA, Unsloth - Kaggle Deep Learning Competitions

Jan 2026 - May 2026

- Fine-tuned SmolVLM-500M-Instruct with rank-16 LoRA adapters on attention projections, training only ~5M parameters, for a visual question answering competition; scored 0.83098 on the public leaderboard within strict model-size and trainable-parameter caps.
- Fine-tuned Qwen2.5-2B with QLoRA using 4-bit quantization (via Unsloth) for text-to-SVG generation, building a validity pipeline (SVG filtering, dimension fixes, minimal-valid fallback) to guarantee constraint-compliant submissions.
- Optimized batched inference to fit a sub-1-hour Colab runtime constraint, and applied letter-shuffle augmentation to counter positional bias plus test-time augmentation over choice-text log-probabilities.

ForkWise: Optimized Inference for a Graph Neural Network

Python, GISMo (GNN), ONNX, Triton, Kubernetes, MLflow, Prometheus

Feb 2026 - May 2026

- Exported a GISMo (Graph-based Ingredient Substitution Module) graph neural network to ONNX and served it via Triton Inference Server for optimized inference, as an MLOps capstone predicting viable recipe substitutes from an ingredient graph.
- Instrumented the deployment with Prometheus to monitor inference latency and throughput in real time, and tracked experiments and model versions with MLflow.
- Containerized and deployed the model-serving pipeline on Chameleon Cloud using Kubernetes, integrating with Mealie, an open-source recipe manager, to serve substitutions within real recipe workflows.

Stock Price Prediction Using RNNs

Python, RNN/LSTM, Time-Series Forecasting

Sep 2025 - Dec 2025

- Built and trained an RNN/LSTM time-series forecasting pipeline on sequential windows of historical price data, evaluating against baseline forecasting approaches to assess how well the model captured temporal dependencies.

SKILLS

Efficient Deep Learning: Parameter-efficient fine-tuning (LoRA/QLoRA), 4-bit quantization, ONNX export, Triton Inference Server, batched inference optimization, training under parameter and runtime budgets

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Transformers, vision-language models, RNN/LSTM, CNN, GNN, adversarial ML, experiment design

Systems & MLOps: Docker, Kubernetes, MLflow, Prometheus, FastAPI, Git, Linux

Languages & Data: Python, Java, JavaScript, SQL, PySpark, Pandas, NumPy, HuggingFace